

EX.NO:1

Basic I/O programming

PROGRAM :

```
#include<stdio.h>
#include<unistd.h>
#include<string.h>
#include<fcntl.h>
main( )
{
int fd[2];
char buf1[25]= "just a test\n";
char buf2[50];
fd[0]=open("file1",O_RDWR);
fd[1]=open("file2",O_RDWR);
write(fd[0], buf1, strlen(buf1));
printf("\n Enter the text now....");
scanf("\n %s",buf1);
printf("\n Cat file1 is \n hai");
write(fd[0], buf1, strlen(buf1));
lseek(fd[0], SEEK_SET, 0);
read(fd[0], buf2, sizeof(buf1));
write(fd[1], buf2, sizeof(buf2));
close(fd[0]);
close(fd[1]);
printf("\n");
return 0;
}
```

OUTPUT:

```
Enter the text now...hai
```

```
Cat file1 Just a
```

```
hai
```

```
Cat file1 is
```

```
hai
```

```
[sree@localhost ~]$ gedit open.c
```

```
[sree@localhost ~]$ cc open.c
```

```
[sree@localhost ~]$ ./a.out
```

```
Enter the text now...hai
```

```
Cat file1 is
```

```
hai
```

```
[sree@localhost ~]$ █
```

EX.NO:2

SHORTEST JOB FIRST SCHEDULING

PROGRAM:

```
#include<stdio.h>
#include<conio.h>
struct process
{
    int pid;
    int bt;
    int wt;
    int tt;
}p[10],temp;
int main()
{
    int i,j,n,totwt,totvt;
    float avg1,avg2;
    clrscr();
    printf("\nEnter the number of process:\t");
    scanf("%d",&n);
    for(i=1;i<=n;i++)
    {
        p[i].pid=i;
        printf("\nEnter the burst time:\t");
        scanf("%d",&p[i].bt);
    }
    for(i=1;i<n;i++)
    {
        for(j=i+1;j<=n;j++)
        {
            if(p[i].bt>p[j].bt)
            {
                temp.pid=p[i].pid;
                p[i].pid=p[j].pid;
                p[j].pid=temp.pid;
            }
        }
    }
}
```

```

        p[j].pid=temp.pid;
        temp.bt=p[i].bt;p[i].bt=p[j].bt;
        p[j].bt=temp.bt;
    }
}
}
p[1].wt=0;
p[1].tt=p[1].bt+p[1].wt;
i=2;
while(i<=n)
{
    p[i].wt=p[i-1].bt+p[i-1].wt;
    p[i].tt=p[i].bt+p[i].wt;
    i++;
}
i=1;
totwt=totwt=0;
printf("\nProcess id \tbt \tw\t\ttt");
while(i<=n)
{
    printf("\n\t%d \t%d \t%d \t%d\n",p[i].pid,p[i].bt,p[i].wt,p[i].tt);
    totwt=p[i].wt+totwt;
    tottt=p[i].tt+tottt;
    i++;}
avg1=totwt/n;
avg2=tottt/n;
printf("\nAVG1=%f\t AVG2=%f",avg1,avg2);
getch();
return 0;
}

```

OUTPUT:

enter the number of process 3

enter the burst time: 2

enter the burst time: 4

enter the burst time: 6

processed	bt	wt	tt
1	2	0	2
2	4	2	6
3	6	6	12

AVG1=2.000000 AVG2=6.000000

EX.NO:3

FIRST COME FIRST SERVE SCHEDULING

PROGRAM :

```
#include<stdio.h>

struct process
{
int pid;
int bt;
int wt,tt;
}p[10];

int main()
{
int i,n,totwt,totwt,avg1,avg2; clrscr();
printf("enter the no of process \n"); scanf("%d",&n);
for(i=1;i<=n;i++)
{ p[i].pid=i;
printf("enter the burst time n"); scanf("%d",&p[i].bt);
}
p[1].wt=0;
p[1].tt=p[1].bt+p[1].wt;
i=2;
while(i<=n)
{
p[i].wt=p[i-1].bt+p[i-1].wt; p[i].tt=p[i].bt+p[i].wt;
i++;
}
i=1;
totwt=totwt=0;
printf("\n processid \t bt\t wt\t tt\n"); while(i<=n)
{
printf("\n\t%d \t%d \t%d \t%d",p[i].pid,p[i].bt,p[i].wt,p[i].tt);
```

```

        totwt=p[i].wt+totwt;
        tottt=p[i].tt+tottt;
        i++;}
    avg1=totwt/n; avg2=tottt/n; printf("\navg1=%d \t avg2=%d
    \t",avg1,avg2); getch();
    return 0;
}

```

OUTPUT:

enter the no of process 3

enter the burst time 2

enter the burst time 4

enter the burst time 6

Processid	bt	wt	tt
1	2	0	2
2	4	2	6
3	6	6	12

avg1=2 avg2=6

EX.NO:4

ROUND ROBIN SCHEDULING ALGORITHM

PROGRAM :

```
#include<stdio.h>
#include<conio.h>
struct process
{
int pid,bt,tt,wt;
};
int main()
{
struct process x[10],p[30];
int i,j,k,tot=0,m,n;
float wtime=0.0,tottime=0.0,a1,a2;
clrscr();
printf("\nEnter the number of process:\t");
scanf("%d",&n);
for(i=1;i<=n;i++){
x[i].pid=i;
printf("\nEnter the Burst Time:\t");
scanf("%d",&x[i].bt);
tot=tot+x[i].bt;
}
printf("\nTotal Burst Time:\t%d",tot);
p[0].tt=0;
k=1;
printf("\nEnter the Time Slice:\t");
scanf("%d",&m);
for(j=1;j<=tot;j++){
for(i=1;i<=n;i++)
```

```

{
if(x[i].bt !=0)
{
p[k].pid=i;
if(x[i].bt-m<0)
{
p[k].wt=p[k-1].tt;
p[k].bt=x[i].bt;
p[k].tt=p[k].wt+x[i].bt;
x[i].bt=0;
k++;
}
else
{
p[k].wt=p[k-1].tt;
p[k].tt=p[k].wt+m;
x[i].bt=x[i].bt-m;
k++;
}
}
}
printf("\nProcess id \tw\t ttt");
for(i=1;i<k;i++){
printf("\n\t%d \t%d \t%d",p[i].pid,p[i].wt,p[i].tt);
wtime=wtime+p[i].wt;
tottime=tottime+p[i].tt;
a1=wtime/n;
a2=tottime/n;
}
printf("\n\nAverage Waiting Time:\t%f",a1);

```

```
printf("\n\nAverage TurnAround Time:\t%f",a2);
getch();
return 0;
}
```

OUTPUT:

enter the no of process3

enter the burst time3

enter the burst time5

enter the burst time7

total burst time : 15

enter the time slice: 2

process id	wt	tt
------------	----	----

1	0	2
---	---	---

2	2	4
---	---	---

3	4	6
---	---	---

1	6	7
---	---	---

process id	wt	tt
------------	----	----

2	7	9
---	---	---

3	9	11
---	---	----

2	11	12
---	----	----

3	12	14
---	----	----

3	14	15
---	----	----

avg waiting time: 21.666666

avg turnaround time: 26.666666

EX.NO:5

PRIORITY SCHEDULING ALGORITHM

PROGRAM:

```
#include<stdio.h>
#include<conio.h>
struct process
{
int pid;
int bt;
int wt;
int tt;
int prior;
}
p[10],temp;
int main()
{
int i,j,n,totwt,totvt,arg1,arg2;
clrscr();
printf("enter the number of process");
scanf("%d",&n);
for(i=1;i<=n;i++)
{
p[i].pid=i;
printf("enter the burst time");
scanf("%d",&p[i].bt);
printf("\n enter the priority");
scanf("%d",&p[i].prior);
}
for(i=1;i<n;i++)
{
for(j=i+1;j<=n;j++)
```

```

{
if(p[i].prior>p[j].prior)
{
temp.pid=p[i].pid;
p[i].pid=p[j].pid;
p[j].pid=temp.pid;
temp.bt=p[i].bt;
p[i].bt=p[j].bt;
p[j].bt=temp.bt;
temp.prior=p[i].prior;
p[i].prior=p[j].prior;
p[j].prior=temp.prior;
}
}
}

p[i].wt=0;
p[1].tt=p[1].bt+p[1].wt;
i=2;
while(i<=n)
{
p[i].wt=p[i-1].bt+p[i-1].wt;
p[i].tt=p[i].bt+p[i].wt;
i++;
}
i=1;
totwt=totwt=0;
printf("\n process to \t bt \t wt \t tt");
while(i<=n)
{

printf("\n%d\t %d\t %d\t %d\t",p[i].pid,p[i].bt,p[i].wt,p[i].tt);

```

```
totwt=p[i].wt+totwt;
tottt=p[i].tt+tottt;
i++;

}
arg1=totwt/n;
arg2=tottt/n;
printf("\n arg1=%d \t arg2=%d\t",arg1,arg2);
getch();
return 0;
}
```

OUTPUT:

enter the no of process:3

enter the burst time:2

enter the priority:3

enter the burst time:4

enter the priority:1

enter the burst time:6

enter the priority:2

process to bt wt tt

1 4 0 4 4

2 6 4 10 14

3 2 10 12 22

avg1=4 avg2=8

EX.NO:6

READER / WRITER PROBLEM USING SEMOPHERES

PROGRAM:

```
#include<semaphore.h>
#include<stdio.h>
#include<pthread.h>

void *reader(void *);
void *writer(void *);

int readcount=0,writecount=0,sh_var=5,bsize[5];
sem_t x,y,z,rsem,wsem;
pthread_t r[3],w[2];

void *reader(void *i)
{
    printf("\n.....");
    printf("\n\n reader-%d is reading",i);

    sem_wait(&z);
    sem_wait(&rsem);
    sem_wait(&x);
    readcount++;
    if(readcount==1)
        sem_wait(&wsem);
    sem_post(&x);
    sem_post(&rsem);
    sem_post(&z);
    printf("\nupdated value : %d",sh_var);
    sem_wait(&x);
```

```
readcount--;  
if(readcount==0)  
    sem_post(&wsem);  
sem_post(&x);  
}
```

```
void *writer(void *i)  
{  
    printf("\n\n writer-%d is writing",i);  
    sem_wait(&y);  
    writecount++;  
    if(writecount==1)  
        sem_wait(&rsem);  
    sem_post(&y);  
    sem_wait(&wsem);
```

```
    sh_var=sh_var+5;  
    sem_post(&wsem);  
    sem_wait(&y);  
    writecount--;  
    if(writecount==0)  
        sem_post(&rsem);  
    sem_post(&y);  
}
```

```
int main()  
{  
    sem_init(&x,0,1);  
    sem_init(&wsem,0,1);  
    sem_init(&y,0,1);  
    sem_init(&z,0,1);
```

```
sem_init(&rsem,0,1);
```

```
pthread_create(&r[0],NULL,(void *)reader,(void *)0);  
pthread_create(&w[0],NULL,(void *)writer,(void *)0);  
pthread_create(&r[1],NULL,(void *)reader,(void *)1);  
pthread_create(&r[2],NULL,(void *)reader,(void *)2);  
pthread_create(&r[3],NULL,(void *)reader,(void *)3);  
pthread_create(&w[1],NULL,(void *)writer,(void *)3);  
pthread_create(&r[4],NULL,(void *)reader,(void *)4);
```

```
pthread_join(r[0],NULL);  
pthread_join(w[0],NULL);  
pthread_join(r[1],NULL);  
pthread_join(r[2],NULL);  
pthread_join(r[3],NULL);  
pthread_join(w[1],NULL);  
pthread_join(r[4],NULL);
```

```
return(0);
```

```
}
```

OUTPUT:

```
student@mmcoe-desktop:~$ gcc rw1.c -lpthread
```

```
student@mmcoe-desktop:~$ ./a.out
```

```
-----
```

```
reader-0 is reading
```

```
updated value : 5
```

```
writer-0 is writing
```

```
-----
```

```
reader-1 is reading
```

```
updated value : 10
```

```
-----
```

```
reader-2 is reading
```

```
updated value : 10
```

```
-----
```

```
reader-3 is reading
```

```
updated value : 10
```

```
writer-3 is writing
```

```
-----
```

```
reader-4 is reading
```

EX.NO:7 BANKER'S ALGORITHM FOR DEADLOCK AVOIDANCE

PROGRAM:

```
#include<stdio.h>
#include<conio.h>
struct da
{
int max[10],a1[10],need[10],before[10],after[10];
}p[10];
void main()
{
int i,j,k,l,r,n,tot[10],av[10],cn=0,cz=0,temp=0,c=0;
clrscr();
printf("\n ENTER THE NO. OF PROCESSES:");
scanf("%d",&n);
printf("\n ENTER THE NO. OF RESOURCES:");
scanf("%d",&r);
for(i=0;i<n;i++)
{
printf("PROCESS %d \n",i+1);
for(j=0;j<r;j++)
{
printf("MAXIMUM VALUE FOR RESOURCE %d:",j+1);
scanf("%d",&p[i].max[j]);
}
for(j=0;j<r;j++)
{
printf("ALLOCATED FROM RESOURCE %d:",j+1);
scanf("%d",&p[i].a1[j]);
p[i].need[j]=p[i].max[j]-p[i].a1[j];
}
}
```

```

}
for(i=0;i<r;i++)
{
printf("ENTER TOTAL VALUE OF RESOURCE %d:",i+1);
scanf("%d",&tot[i]);
}
for(i=0;i<r;i++)
{
for(j=0;j<n;j++)
temp=temp+p[j].a1[i];
av[i]=tot[i]-temp;
temp=0;
}
printf("\n\t RESOURCES ALLOCATED NEEDED TOTAL AVAIL");
for(i=0;i<n;i++)
{
printf("\n P%d \t",i+1);
for(j=0;j<r;j++)
printf("%d",p[i].max[j]);
printf("\t");
for(j=0;j<r;j++)
printf("%d",p[i].a1[j]);
printf("\t");
for(j=0;j<r;j++)
printf("%d",p[i].need[j]);
printf("\t");
for(j=0;j<r;j++)
{
if(i==0)
printf("%d",tot[j]);
}
}

```

```

printf(" ");
for(j=0;j<r;j++)
{
if(i==0)
printf("%d",av[j]);
}
}
printf("\n\n\t AVAIL BEFORE\t AVAIL AFTER ");
for(l=0;l<n;l++)
{
for(i=0;i<n;i++)
{
for(j=0;j<r;j++)
{
if(p[i].need[j] > av[j])
cn++;
if(p[i].max[j]==0)
cz++;
}
if(cn==0 && cz!=r)
{
for(j=0;j<r;j++)
{
p[i].before[j]=av[j]-p[i].need[j];
p[i].after[j]=p[i].before[j]+p[i].max[j];
av[j]=p[i].after[j];
p[i].max[j]=0;
}
printf("\n P %d \t",i+1);
for(j=0;j<r;j++)
printf("%d",p[i].before[j]);

```

```
printf("\t");
for(j=0;j<r;j++)
printf("%d",p[i].after[j]);
cn=0;
cz=0;
c++;
break;
}
else
{
cn=0;cz=0;
}
}
}
if(c==n)
printf("\n THE ABOVE SEQUENCE IS A SAFE SEQUENCE");
else
printf("\n DEADLOCK OCCURED");
getch();
}
```

OUTPUT:

Expected Input and Output:

ENTER THE NO. OF PROCESSES:5

ENTER THE NO. OF RESOURCES:3

PROCESS 1

MAXIMUM VALUE FOR RESOURCE 1:7

MAXIMUM VALUE FOR RESOURCE 2:5

MAXIMUM VALUE FOR RESOURCE 3:3

ALLOCATED FROM RESOURCE 1:0

ALLOCATED FROM RESOURCE 2:1

ALLOCATED FROM RESOURCE 3:0

PROCESS 2

MAXIMUM VALUE FOR RESOURCE 1:3

MAXIMUM VALUE FOR RESOURCE 2:2

MAXIMUM VALUE FOR RESOURCE 3:2

ALLOCATED FROM RESOURCE 1:2

ALLOCATED FROM RESOURCE 2:0

ALLOCATED FROM RESOURCE 3:0

PROCESS 3

MAXIMUM VALUE FOR RESOURCE 1:9

MAXIMUM VALUE FOR RESOURCE 2:0

MAXIMUM VALUE FOR RESOURCE 3:2

ALLOCATED FROM RESOURCE 1:3

ALLOCATED FROM RESOURCE 2:0

ALLOCATED FROM RESOURCE 3:2

PROCESS 4

MAXIMUM VALUE FOR RESOURCE 1:2

MAXIMUM VALUE FOR RESOURCE 2:2

MAXIMUM VALUE FOR RESOURCE 3:2

ALLOCATED FROM RESOURCE 1:2

ALLOCATED FROM RESOURCE 2:1

ALLOCATED FROM RESOURCE 3:1

PROCESS 5

MAXIMUM VALUE FOR RESOURCE 1:4

MAXIMUM VALUE FOR RESOURCE 2:3

MAXIMUM VALUE FOR RESOURCE 3:3

ALLOCATED FROM RESOURCE 1:0

ALLOCATED FROM RESOURCE 2:0

ALLOCATED FROM RESOURCE 3:2

ENTER TOTAL VALUE OF RESOURCE 1:10

ENTER TOTAL VALUE OF RESOURCE 2:5

ENTER TOTAL VALUE OF RESOURCE 3:7

RESOURCES ALLOCATED NEEDED TOTAL AVAIL

P1 753 010 743 1057 332

P2 322 200 122

P3 902 302 600

P4 222 211 011

P5 433 002 431

AVAIL BEFORET AVAIL AFTER

P 2 210 532

P 4 521 743

P 1 000 753

P 3 153 1055

P 5 624 1057

THE ABOVE SEQUENCE IS A SAFE SEQUENCE

EX NO: 8

FIFO PAGE REPLACEMENT ALGORITHMS

PROGRAM:

```
#include<stdio.h>
#include<conio.h>
int i,j,nof,nor,flag=0,ref[50],frm[50],pf=0,victim=-1;
void main()
{
clrscr();
printf("\n \t\t\t FIFI PAGE REPLACEMENT ALGORITHM");
printf("\n Enter no.of frames....");
scanf("%d",&nof);
printf("Enter number of Pages.\n");
scanf("%d",&nor);
printf("\n Enter the Page No.  ");
for(i=0;i<nor;i++)
scanf("%d",&ref[i]);
printf("\nThe given Pages are:");
for(i=0;i<nor;i++)
printf("%4d",ref[i]);
for(i=1;i<=nof;i++)
frm[i]=-1;
printf("\n");
for(i=0;i<nor;i++)
{
flag=0;
printf("\n\t page no %d->\t",ref[i]);
for(j=0;j<nof;j++)
{
if(frm[j]==ref[i])
{
```

```
flag=1;
break;
}
}
if(flag==0)
{
pf++;
victim++;
victim=victim%nof;
frm[victim]=ref[i];
for(j=0;j<nof;j++)
printf("%4d",frm[j]);
}
}
printf("\n\n\t\t No.of pages faults...%d",pf);
getch();
}
```

OUTPUT:**FIFO PAGE REPLACEMENT ALGORITHM**

Enter no.of frames....

4

Enter number of reference string..

6

Enter the reference string..

5 6 4 1 2 3

The given reference string:

..... 5 6 4 1 2 3

Reference np5-> 5 -1 -1 -1

Reference np6-> 5 6 -1 -1

Reference np4-> 5 6 4 -1

Reference np1-> 5 6 4 1

Reference np2-> 2 6 4 1

Reference np3-> 2 3 4 1

No.of pages faults ..6

EX NO:9 LRU PAGE REPLACEMENT ALGORITHM

PROGRAM:

```
#include<stdio.h>
#include<conio.h>
int i,j,nof,nor,flag=0,ref[50],frm[50],pf=0,victim=-1;
int recent[10],lrucal[50],count=0;
int lruvictim();
void main()
{
clrscr();
printf("\n\t\t\t LRU PAGE REPLACEMENT ALGORITHM");
printf("\n Enter no.of Frames... ");
scanf("%d",&nof);
printf(" Enter no.of reference string..");
scanf("%d",&nor);
printf("\n Enter reference string..");
for(i=0;i<nor;i++)
scanf("%d",&ref[i]);
printf("\n\n\t\t\t LRU PAGE REPLACEMENT ALGORITHM ");
printf("\n\t\t The given reference string:");
printf("\n ..... ");
for(i=0;i<nor;i++)
printf("%4d",ref[i]);
for(i=1;i<=nof;i++)
{
frm[i]=-1;
lrucal[i]=0;
}
for(i=0;i<10;i++)
recent[i]=0;
```

```
printf("\n");
for(i=0;i<nor;i++)
{
flag=0;
printf("\n\t Reference NO %d->\t",ref[i]);
for(j=0;j<nof;j++)
{
if(frm[j]==ref[i])
{
flag=1;
break;
}
}
if(flag==0)
{
count++;
if(count<=nof)
victim++;
else
victim=lruvictim();
pf++;
frm[victim]=ref[i];
for(j=0;j<nof;j++)
printf("%4d",frm[j]);
}
recent[ref[i]]=i;
}
printf("\n\n\t No.of page faults...%d",pf);
getch();
}
int lruvictim()
```

```
{
int i,j,temp1,temp2;
for(i=0;i<nof;i++)
{
temp1=frm[i];
lrucal[i]=recent[temp1];
}
temp2=lrucal[0];
for(j=1;j<nof;j++)
{
if(temp2>lrucal[j])
temp2=lrucal[j];
}
for(i=0;i<nof;i++)
if(ref[temp2]==frm[i])
return i;
return 0;
}
```

OUTPUT:

LRU PAGE REPLACEMENT ALGORITHM

Enter no.of Frames ... 3

Enter no.of reference string. 6

Enter reference string..

6 5 4 2 3 1

LRU PAGE REPLACEMENT ALGORITHM

The given reference string:

..... 6 5 4 2 3 1

Reference NO 6-> 6 -1 -1

Reference NO 5-> 6 5 -1

Reference NO 4-> 6 5 4

Reference NO 2-> 2 5 4

Reference NO 3-> 2 3 4

Reference NO 1-> 2 3 1

No.of page faults..6

EX.NO:10**OPTIMAL PAGE REPLACEMENT ALGORITHM****PROGRAM:**

```
#include<stdio.h>
#include<conio.h>
int i,j,nof,nor,flag=0,ref[50],frm[50],pf=0,victim=-1;
int recent[10],optcal[50],count=0;
int optvictim();
void main()
{
clrscr();
printf("\n OPTIMAL PAGE REPLACEMENT ALGORITHM");
printf("\n ..... ");
printf("\nEnter the no.of frames");
scanf("%d",&nof);
printf("Enter the no.of reference string");
scanf("%d",&nor);
printf("Enter the reference string");
for(i=0;i<nor;i++)
scanf("%d",&ref[i]);
clrscr();
printf("\n OPTIMAL PAGE REPLACEMENT ALGORITHM");
printf("\n ..... ");
printf("\nThe given string");
printf("\n ..... \n");
for(i=0;i<nor;i++)
printf("%4d",ref[i]);
for(i=0;i<nof;i++)
{
frm[i]=-1;
optcal[i]=0;
```

```

}
for(i=0;i<10;i++)
recent[i]=0;
printf("\n");
for(i=0;i<nor;i++)
{
flag=0;
printf("\n\tref no %d ->\t",ref[i]);
for(j=0;j<nof;j++)
{
if(frm[j]==ref[i])
{
flag=1;
break;
}
}
if(flag==0)
{
count++;
if(count<=nof)
victim++;
else
victim=optvictim(i);
pf++;
frm[victim]=ref[i];
for(j=0;j<nof;j++)
printf("%4d",frm[j]);
}

}
printf("\n Number of page faults: %d",pf);

```

```
getch();
}
int optvictim(int index)
{
int i,j,temp,notfound;
for(i=0;i<nof;i++)
{
notfound=1;
for(j=index;j<nor;j++)
if(frm[i]==ref[j])
{
notfound=0;
optcal[i]=j;
break;
}
if(notfound==1)
return i;
}
temp=optcal[0];
for(i=1;i<nof;i++)
if(temp<optcal[i])
temp=optcal[i];
for(i=0;i<nof;i++)
if(frm[temp]==frm[i])
return i;
return 0;
}
```

OUTPUT:

OPTIMAL PAGE REPLACEMENT ALGORITHM

Enter no. of Frames... 3

Enter no. of reference string. 6

Enter reference string..

6 5 4 2 3 1

OPTIMAL PAGE REPLACEMENT ALGORITHM

The given reference string:

..... 6 5 4 2 3 1

Reference NO 6-> 6 -1 -1

Reference NO 5-> 6 5 -1

Reference NO 4-> 6 5 4

Reference NO 2-> 2 5 4

Reference NO 3-> 2 3 4

Reference NO 1-> 2 3 1

No.of page faults.. 6

EX.NO:11

FIRST BIT ALGORITHM FOR MEMORY MANAGEMENT

PROGRAM:

```
#include<stdio.h>
#include<conio.h>
#define max 25
void main()
{
int frag[max],b[max],f[max],i,j,nb,nf,temp;
static int bf[max],ff[max];
clrscr();
printf("\n\tMemory Management Scheme - First Fit");
printf("\n\tEnter the number of blocks:");
scanf("%d",&nb);
printf("Enter the number of files:");
scanf("%d",&nf);
printf("\n\tEnter the size of the blocks:-\n");
for(i=1;i<=nb;i++)
{
printf("Block %d:",i);
scanf("%d",&b[i]);
}
printf("Enter the size of the files :-\n");
for(i=1;i<=nf;i++)
{
printf("File %d:",i);
scanf("%d",&f[i]);
}
for(i=1;i<=nf;i++)
{
for(j=1;j<=nb;j++)
```

```
{
if(bf[j]!=1)
{
temp=b[j]-f[i];
if(temp>=0)
{
ff[i]=j;
break;
}
}
}
frag[i]=temp;
bf[ff[i]]=1;
}
printf("\nFile_no:\tFile_size :\tBlock_no:\tBlock_size:\tFragement");
for(i=1;i<=nf;i++)
printf("\n%d\t\t%d\t\t%d\t\t%d\t\t%d",i,f[i],ff[i],b[ff[i]],frag[i]);
getch();
}
```

OUTPUT:

INPUT

Enter the number of blocks: 3

Enter the number of files: 2

Enter the size of the blocks:-

Block 1: 5

Block 2: 2

Block 3: 7

Enter the size of the files:-

File 1: 1

File 2: 4

OUTPUT

File No	File Size	Block No	Block Size	Fragment
1	1	1	5	4
2	4	3	7	3

EX.NO:12

BEST BIT ALGORITHM FOR MEMORY MANAGEMENT

PROGRAM:

```
#include<stdio.h>
#include<conio.h>
#define max 25
void main()
{
int frag[max],b[max],f[max],i,j,nb,nf,temp,lowest=10000;
static int bf[max],ff[max];
clrscr();
printf("\nEnter the number of blocks:");
scanf("%d",&nb);
printf("Enter the number of files:");
scanf("%d",&nf);
printf("\nEnter the size of the blocks:-\n");
for(i=1;i<=nb;i++)
{
printf("Block %d:",i);
scanf("%d",&b[i]);
}
printf("Enter the size of the files :-\n");
for(i=1;i<=nf;i++)
{
printf("File %d:",i);
scanf("%d",&f[i]);
}
for(i=1;i<=nf;i++)
{
for(j=1;j<=nb;j++)
```

```
{
if(bf[j]!=1)
{
temp=b[j]-f[i];
if(temp>=0)
if(lowest>temp)
{
ff[i]=j;

lowest=temp;
}
}
}
frag[i]=lowest;
bf[ff[i]]=1;
lowest=10000;
}
printf("\nFile No\tFile Size \tBlock No\tBlock Size\tFragment");
for(i=1;i<=nf && ff[i]!=0;i++)
printf("\n%d\t\t%d\t\t%d\t\t%d\t\t%d",i,f[i],ff[i],b[ff[i]],frag[i]);
getch();
}
```

OUTPUT:

INPUT

Enter the number of blocks: 3

Enter the number of files: 2

Enter the size of the blocks:-

Block 1: 5

Block 2: 2

Block 3: 7

Enter the size of the files:-

File 1: 1

File 2: 4

OUTPUT

File No	File Size	Block No	Block Size	Fragment
1	1	2	2	1
2	4	1	5	1

EX.NO:13

WORST BIT ALGORITHM FOR MEMORY MANAGEMENT

PROGRAM:

```
#include<stdio.h>
#include<conio.h>
#define max 25
void main()
{
int frag[max],b[max],f[max],i,j,nb,nf,temp,highest=0;
static int bf[max],ff[max];
clrscr();
printf("\n\tMemory Management Scheme - Worst Fit");
printf("\nEnter the number of blocks:");
scanf("%d",&nb);
printf("Enter the number of files:");
scanf("%d",&nf);
printf("\nEnter the size of the blocks:-\n");
for(i=1;i<=nb;i++)
{
printf("Block %d:",i);
scanf("%d",&b[i]);
}
printf("Enter the size of the files :-\n");
for(i=1;i<=nf;i++)
{
printf("File %d:",i);
scanf("%d",&f[i]);
}
for(i=1;i<=nf;i++)
{
```

```
for(j=1;j<=nb;j++)
{
if(bf[j]!=1) //if bf[j] is not allocated
{
temp=b[j]-f[i];
if(temp>=0)
if(highest<temp)
{
ff[i]=j;
highest=temp;
}
}
}
frag[i]=highest;
bf[ff[i]]=1;
highest=0;
}
printf("\nFile_no:\tFile_size :\tBlock_no:\tBlock_size:\tFragement");
for(i=1;i<=nf;i++)
printf("\n%d\t\t%d\t\t%d\t\t%d\t\t%d",i,f[i],ff[i],b[ff[i]],frag[i]);
getch();
}
```

OUTPUT:

INPUT

Enter the number of blocks: 3

Enter the number of files: 2

Enter the size of the blocks:-

Block 1: 5

Block 2: 2

Block 3: 7

Enter the size of the files:-

File 1: 1

File 2: 4

OUTPUT

File No	File Size	Block No	Block Size	Fragment
1	1	3	7	6
2	4	1	5	1

EX.NO:14**INTERPROCESS COMMUNICATION USING PIPES****PROGRAM:**

```
#include<stdio.h>
#include<unistd.h>
#include<string.h>
main()
{
    int p1[2],p2[2],p3[2],p4[2];
    int i,j=0,k=0,l=0;
    char r[10],s[10],t[10],u[10];
    printf("\t PROCESS 1. ENTER THE STRING");
    scanf("%s",r);
    pipe(p1);
    pipe(p2);
    write(p1[1],r,sizeof(r));
    write(p2[1],r,sizeof(r));
    int a=fork();
    if(a==0)
    {
        printf("\n\t PROCESS 2: it splits the given string\n");
        read(p1[0],r,sizeof(r));
        int n=strlen(r);
        for(i=0;i<n/2;i++)
        {
            s[i]=r[i];
        }
        for(i=n/2;i<=n;i++)
        {
            t[j++]=r[i];
        }
    }
```

```
pipe(p3);
pipe(p4);
write(p3[1],s,sizeof(s));
write(p4[1],t,sizeof(t));
int b=fork();
if(b==0)
{
printf("p4 %d\t",getpid());
printf("p2 %d\n",getppid());
read(p3[0],s,sizeof(s));
printf("\t PROCESS 4:sub string \t %s \t",s);
printf("no of char=%d \n",strlen(s));
}
else
{
int c=fork();
if(c==0)
{
printf("p5 %d\t",getpid());
printf("p2 %d\n",getppid());
read(p4[0],t,sizeof(t));
printf("\t PROCESS 5:sub string \t %s \t",t);
printf("no of char=%d \n",strlen(t));
}
else
{
wait();
printf("p2 %d\t",getpid());
printf("p1 %d\n",getppid());
} }
}
```

```

{
    wait();
    int d=fork();
    if(d==0)
    {
        printf("p3 %d\t",getpid());
        printf("p1 %d\n",getppid());
        read(p2[0],r,sizeof(r));
        for(i=strlen(r)-1;i>=0;i--)
        {
            u[l++]=r[i];
        }
        for(i=0;i<strlen(r);i++)
        {
            if(u[i]==r[i])
                k++;
            else
                continue;
        }
        if(k==strlen(r))
            printf("\t PROCESS 3: the given string is palindrome\n");
        else
            printf("\t PROCESS 3: the given string is not palindrome\n");
    }
    else
    {
        printf("p1 %d\t",getpid());
        printf("kernal %d\t\n",getppid());
    }
}}

```

OUTPUT:

Process 1: enter the string ARUN

Process 2: it splits the string

P4 8137 p2 8136

Process 3: sub string a r no of char=2

P5 8138 p2 8136

Process 4; substring u n no of char=2

P2 8136 p1 8132

P3 8139 p1 8132

Process 3: the given string is not palindrome

Process 1: enter the string MADAM

Process 2: it splits the string

P4 8137 p2 8136

Process 4: sub string m a no of char=2

P5 8138 p2 8136

Process 5: sub string dam no of char=3

P2 8136 p1 8132

P3 8139 p1 8132

Process 3: the given string is palindrome